

VEDANT PATIL

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SUMMARY

Software Engineer with a DevOps and platform background, shipping production **Kubernetes** across **AWS** and air-gapped edge with **CI/CD**, **GitOps**, and observability, now extended to production AI: **MCP** servers, **AIOps** agents, and **GPU-scale inference**.

SKILLS

Cloud & Infrastructure: AWS (EKS, ECS Fargate, VPC, IAM, CloudWatch, Bedrock), Azure, Terraform, Cloudflare

Containers & Orchestration: Kubernetes (K8s, K3s, EKS), Docker, Docker Compose, Helm, NVIDIA Container Toolkit

CI/CD & GitOps: GitHub Actions, Jenkins, ArgoCD, automated security scanning

Observability & Monitoring: Prometheus, Grafana, Loki, Promtail, CloudWatch, Distributed Tracing, Auto-Scaling, Cost Governance

Networking & Edge: VXLAN Overlay, CNI configuration, pod-to-pod routing, air-gapped deployment strategies

Security & Compliance: HIPAA, Secrets Management, IAM / RBAC, Multi-stage and Distrosless Container Builds

Languages & Scripting: Python, Bash, JavaScript / TypeScript, C++

Backends & Data: FastAPI, Node.js, PostgreSQL, MongoDB, Redis, REST APIs, MQTT

Agentic AI & LLM Systems: MCP Server, Multi-Agent Orchestration, Claude Code, Codex, n8n, Agentic RAG

WORK EXPERIENCE

Research Associate | ReRout Laboratory, The Ohio State University

December 2024 - May 2026

- Architected and shipped a production **Kubernetes** AI inference platform for **air-gapped edge** environments on NVIDIA Jetson, achieving **96% pipeline success rate** in fully disconnected deployments.
- Resolved a production CNI failure by configuring a **VXLAN overlay** that enabled pod-to-pod traffic without modifying the host network.
- Built air-gapped deployment workflows on **Docker Compose** and **K3s** that eliminated runtime dependencies and cut bootstrap time by **68x**.
- Instrumented heterogeneous edge hardware with **Grafana, Loki, Promtail** for end-to-end observability, and engineered fault-tolerant **MQTT**-driven microservices with stage-dependent error recovery.

Cloud & DevOps Engineer Intern | Healiom

April 2024 - July 2024

- Designed and delivered shared **AWS** infrastructure (EKS, ECS, VPC) using **Terraform**, serving **4 cross-functional teams** in a **HIPAA-regulated** environment.
- Built **4 GitHub Actions CI/CD pipelines** with automated security scanning, secrets management, and **IAM/RBAC** policies.
- Introduced **auto-scaling governance** across production workloads that maintained infrastructure costs within allocated budgets.
- Reduced container image size by **65%** (5 GB to 1.5 GB) through **multi-stage and distrosless builds**, cutting storage and transfer costs.

PROJECTS

RetailOps / Kira | “Autonomous AIOps incident-response agent on AWS with full GitOps”

- Built a full microservices platform on **EKS** provisioned with **Terraform** and deployed through **ArgoCD** and GitHub Actions GitOps workflows.
- Engineered an autonomous **AIOps** agent correlating **CloudWatch** logs, Prometheus metrics, and EKS cluster health in real time to surface root causes without human triage.
- Integrated Claude Code with **AWS MCP servers** for natural-language infrastructure operations across Terraform, EKS, and cost analysis, backed by a **Prometheus/Grafana** stack.

WorldForge | “Production-scale 14B-parameter GPU inference platform on Kubernetes”

- Deployed **NVIDIA Cosmos-Predict2.5-14B** on **K3s** with NVIDIA Container Toolkit, **H100** GPU orchestration, Redis-backed async queues, and FastAPI microservices.
- Authored **23 Kubernetes manifests** enabling **zero-downtime deployments**, achieving **99.9% uptime** and ~90% GPU utilization.
- Delivered **198s P95 latency** at ~\$0.11 cost-per-video and ~18 videos/hour throughput with real-time GPU telemetry and alerting.

Autonomous Multi-Agent Dev Team | “Fully autonomous software development via role-based AI agents”

- Built a fully autonomous **multi-agent** software-engineering pipeline with **role-based agents** (architect, coder, reviewer, debugger) that plan, write, review, and test code with no runtime human intervention.
- Orchestrated inter-agent collaboration through shared context and **MCP tool integrations** with **Codex** and built evaluation harnesses measuring task completion rate and a **failure-mode taxonomy**.

DroneMCP | “Autonomous drone operations platform on AWS ECS Fargate”

- Deployed a production Python MCP server on **AWS ECS Fargate** exposing **45 LLM-callable tools** for flight control, GPS, camera, and ML inference.
- Architected a cloud-to-edge bridge using FastMCP with SSE transport and **ALB-backed WebSocket** routing, with **distributed tracing** across the full bridge.
- Verified deterministic behavior across all tool paths with **35-unit tests** and mocked SDKs before hardware integration.

EDUCATION

Master of Science, Computer Science and Engineering | The Ohio State University

May 2026

Bachelor of Engineering, Information Technology | University of Mumbai

May 2024